# 34. Review 2024 Exam 2

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## 3. Uniform or Importance Sampling (20 pt)

We want to estimate

$$\int_0^1 g(t)dt \quad \text{with} \quad g(t) = \frac{1}{\sqrt{1+t^2}}.$$

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Uniform}[0,1]$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} f(t)$ where

$$f(t) = \frac{1}{\log(2)} \frac{1}{2-t}$$

on $[0,1]$. Determine which of $n^{-1}\sum_{i=1}^n g(X_i)$ and $n^{-1}\sum_{i=1}^n g(Y_i)/f(Y_i)$ has the smaller variance.[1]

Note: Both cases are unbiased estimators.

The difference in the variances is just a difference in the 2nd moments

$$Var(X) = \boxed{\mathbb{E}X^2} - (\mathbb{E}X)^2$$

Recall $Var\left(\sum g(X_i)\right) = \sum Var(g(X_i))$

$= n\, Var(g(X))$

$$\underbrace{Var\left(\frac{1}{n}\sum g(X_i)\right)}_{\frac{1}{n} Var(g(X))} - Var\left(\frac{1}{n}\sum \frac{g(Y_i)}{f(Y_i)}\right)$$

$$\mathbb{E}\left[g(X)^2\right] = \int_0^1 \frac{1}{1+t^2}\,dt = \arctan(t)\Big|_0^1 = \arctan(1) = \pi/4$$

$$\mathbb{E}\left[\frac{g(Y)^2}{f(Y)^2}\right] = \int_0^1 \frac{g(t)^2}{f(t)^2} f(t)\,dt = \int_0^1 \frac{g(t)^2}{f(t)}\,dt$$

$$= \int_0^1 \frac{1}{1+t^2} \cdot (\log 2)(2-t)\,dt$$

$$\frac{1}{2}\log(1+t^2)\Big|_0^1$$

$$= 2\log 2 \int_0^1 \frac{1}{1+t}\,dt - 2\log 2 \int_0^1 \frac{t}{1+t^2}\,dt$$

$$= \frac{\pi}{2}\log 2 - (2\log 2)\left(\frac{1}{2}\log 2\right)$$

$$= \frac{\pi}{2}\log 2 - (\log 2)^2$$

## 4. One more Jackknife Question (20 pt)

Let $U_1, \ldots, U_n \overset{\text{iid}}{\sim} \text{Uniform}[0,b]$ and let the estimator for $b$ and the leave-one-out estimator be, respectively,

$$\hat{b}^{(n)} = \left(\frac{n+2}{n+1}\right) \max_{i=1,\ldots,n}\{U_i\} \quad \text{and} \quad \hat{b}_j^{(n-1)} = \left(\frac{n+1}{n}\right) \max_{i=1,\ldots,j-1,j+1,\ldots,n}\{U_i\}.$$

leave one out

Compute the biases of $\hat{b}$ and the jackknife debiased estimator

$$\boxed{n\hat{b}^{(n)} - \left(\frac{n-1}{n}\right)\sum_{j=1}^n \hat{b}_j^{(n-1)}}$$

to determine which has the smaller bias.

Hint: $\boxed{\mathbb{E}\left(\max_{i=1,\ldots,n}\{U_i\}\right) = nb/(n+1).}$

do the leave one out estimator and use it to estimate/reduce bias.

to determine which has the smaller bias.
Hint: $\mathrm{E}\left(\max_{i=1,\ldots,n}\{U_i\}\right) = nb/(n+1)$.

$$\mathrm{Bias}(\hat{b}) = \mathbb{E}\hat{b} - b$$

$$\mathbb{E}\hat{b} = \mathbb{E}\left[\left(\frac{n+2}{n+1}\right)\max_{1\ldots n}\{U_i\}\right] = \left(\frac{n+2}{n+1}\right)\frac{n}{n+1}\,b$$

$$= \frac{(n+2)n}{(n+1)^2}\,b - b = \frac{\cancel{n^2} + \cancel{2n} - (\cancel{n^2} + \cancel{2n} + 1)}{(n+1)^2}\,b = \boxed{\frac{-1}{(n+1)^2}\,b}$$

$$\mathrm{Bias}\left(n\hat{b}^{(n)} - \left(\frac{n-1}{n}\right)\sum b_j^{(n-1)}\right)$$

$$\mathbb{E}[\ \ ] = \frac{n^2}{n+1}\,b - \left(\frac{n-1}{n}\right)\sum_{j=1}^{n}\left[\frac{n-1}{n}\,b\right]$$

$$= \underbrace{b\left[\frac{n^2}{n+1} - \frac{(n-1)^2}{n}\right]}_{\mathbb{E}[\ ]} - b$$

$$\mathbb{E}[\ ] - b$$

## 5. Stochastic Moments (20 pt)

Do what I did in class with $\bar{X}$

Let $X_1, X_2, \ldots$ be iid random variables with finite variance. The estimator for the first and second[2] moment are

$$\begin{pmatrix}\mu_n\\ \sigma_n^2\end{pmatrix} = \frac{1}{n}\sum_{i=1}^{n}\begin{pmatrix}X_i\\ X_i^2\end{pmatrix}.$$

Write this estimator in the form of stochastic gradient descent, i.e. find $a_n$ and $y(\ )$ such that

$$\begin{pmatrix}\mu_{n+1}\\ \sigma_{n+1}^2\end{pmatrix} = \begin{pmatrix}\mu_n\\ \sigma_n^2\end{pmatrix} - a_n y(\mu_n, \sigma_n^2, X_{n+1}).$$

← Find this piece

$$\begin{pmatrix}\mu_n\\ \sigma_n^2\end{pmatrix} = \begin{pmatrix}\frac{1}{n}\sum X_i\\ \frac{1}{n}\sum X_i^2\end{pmatrix}$$

$$a_n\, y(\ ) = \begin{pmatrix}\mu_{n+1}\\ \sigma_{n+1}^2\end{pmatrix} - \begin{pmatrix}\mu_n\\ \sigma_n^2\end{pmatrix}$$

$$= \begin{pmatrix}\frac{1}{n+1}\sum_{i=1}^{n+1} X_i - \frac{1}{n}\sum X_i\\ \frac{1}{n+1}\sum X_i^2 - \frac{1}{n}\sum X_i^2\end{pmatrix}$$

$$\frac{1}{n+1} - \frac{1}{n} = \frac{n-n-1}{n(n+1)} = \frac{-1}{n(n+1)}$$

$$\frac{1}{n+1}X_{n+1} + \underbrace{\frac{1}{n+1}\sum_{i=1}^{n} X_i - \frac{1}{n}\sum_{i=1}^{n} X_i}$$

$$\frac{1}{n+1} x_{n+1} - \frac{1}{n+1} \sum_{i=1} x_i \qquad n \sum_{i=1} x_i$$

$$\frac{1}{n+1} x_{n+1} - \frac{1}{n(n+1)} \sum x_i$$

$$\frac{1}{n+1} \left( x_{n+1} - \mu_n \right)$$

$$\frac{1}{n+1} \left( x_{n+1} - \mu_n \right)$$

$$\frac{1}{n+1} \left( x_{n+1}^2 - \sigma_n^2 \right)$$

$a_n$

$y(\quad)$